Memory or misspecification? When does a fractional-order epidemic model measure real memory?

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Abstract

Fractional-order epidemic models replace the ordinary time derivative with a Caputo derivative of order $\alpha \in (0, 1]$, and fitted values $\alpha < 1$ are routinely read as evidence that an epidemic has "memory". Yet slow, sub-exponential epidemic growth also arises in integer-order models with gamma-distributed stage durations, time-varying transmission or delayed reporting. We ask when a fitted fractional order measures genuine memory and when it merely absorbs model misspecification. We study a dimensionally consistent Caputo fractional SEIR model and the fractional-recovery SIR model of Angstmann, Henry and McGann, which is derived from heavy-tailed infectious periods; establish existence, uniqueness, positivity and stability results; and characterise what case data can and cannot identify. Using simulation from memoryless models and public data on mpox (United States, 2022), Ebola (Guinea, Liberia and Sierra Leone, 2014) and three US measles outbreaks (2025–26), we test whether fractional fits report spurious memory, profile the likelihood of α, and compare models by AIC and rolling-origin forecasts. *[Results to be added in Stages 4–6.]*

Keywords: fractional calculus; Caputo derivative; epidemic models; identifiability; model misspecification; Ebola; mpox; measles

1 Introduction

In a standard compartmental epidemic model the rate of change of each compartment depends only on the present state of the system: the model has no memory. Fractional-order models change this by replacing $\mathrm{d}/\mathrm{d}t$ with a Caputo derivative of order $\alpha \in (0, 1]$, which is a weighted average of the whole past rate of change, with weights that decay like a power of the elapsed time [10, 4]. When $\alpha = 1$ the ordinary model is recovered; when $\alpha < 1$ the past keeps influencing the present, and growth is slower than exponential.

Two difficulties make a fitted α hard to interpret. First, many fractional epidemic models simply place α in front of the ordinary equations, so that the two sides of each equation have different units; Diethelm [5] showed how to avoid this by raising each rate constant to the power α. Second, slow growth is not unique to memory. Early epidemic growth is often sub-exponential [12, 3], and the same pattern is produced by integer-order mechanisms: gamma-distributed (multi-stage) latent and infectious periods, transmission that declines through behaviour change or interventions, and delays between infection and report. If an integer-order process of this kind generates the data, a fractional model may still fit well, and report $\alpha < 1$, because α is the only flexible parameter it has.

A principled alternative derives the fractional model from a mechanism. Angstmann, Henry and McGann [1] obtained a fractional-recovery SIR model from a stochastic process in which infectious periods have a heavy-tailed (power-law) distribution. In that model α has a direct meaning: it is the tail exponent of the infectious-period distribution.

This paper asks: *when does a fitted fractional order measure real memory, and when does it absorb misspecification?* Our planned contributions are:

1. analysis of a dimensionally consistent Caputo SEIR model and of the model of Angstmann, Henry and McGann (existence, uniqueness, positivity, reproduction number, stability by Matignon's criterion [8]), together with a proposition on what case data can and cannot identify (the order α versus transmission);
2. a validated numerical solver (Garrappa's predictor–corrector or the L1 scheme [6]), with a convergence test against an exact Mittag-Leffler solution;
3. simulation experiments in which data are generated by memoryless models and fitted with the fractional model, to measure how often a spurious $\alpha < 1$ is reported;
4. fits to seven public outbreak series with profile likelihoods for α, AIC model comparison and rolling-origin forecasts.

Section 2 describes the data. Sections 3–8 are outlined and will be completed in later stages.

2 Data

All data are public and are downloaded from GitHub. To make every number in this paper exactly reproducible, each file is downloaded from a fixed commit (a frozen snapshot of its repository) rather than from the repository's current version, and its SHA-256 fingerprint is checked after download (Table 1). Every step below is carried out in Python by the scripts listed in Appendix A.

Table 1: Data sources. Each file is downloaded from the commit shown and verified by its SHA-256 fingerprint.

Dataset	Repository and file	Commit	Content
Mpox, US	`owid/monkeypox`: `owid-monkeypox-data.csv`	`08f97c1`	WHO confirmed cases [9]
Ebola	`cmrivers/ebola`: `country_timeseries.csv`	`ddcbf44`	cumulative cases [11]
Measles	`CSSEGISandData/measles_data`: county file and `Top_states_time_series.csv`	`c361b38`	lab-confirmed cases [7]

2.1 Mpox, United States, 2022

Our World in Data (OWID) publishes the World Health Organization's count of confirmed mpox cases as a daily file, but United States counts arrive once a week: the daily column of new cases is zero on six days and holds the whole week's count on the seventh, always a Monday. We therefore build the weekly series from the running total $C_{\mathrm{OWID}}(t)$, as

$$y_k = C_{\mathrm{OWID}}(m_k) - C_{\mathrm{OWID}}(m_{k-1}), \tag{1}$$

where m_k is the k-th Monday, and label week k by m_k. The running total never decreases, so $y_k \geq 0$ for every k.

The 2022 wave is taken to run from the first report (9 May 2022) to the first week after the peak whose count falls below 2% of the peak. This objective rule avoids choosing the end date by eye. The peak is 3,270 cases (week of 1 August 2022); 2% of the peak is 65.4 cases, first undershot in the week of 19 December 2022 (62 cases). The resulting series has 33 weeks and 30,056 cases. As an independent check, the weekly counts were recomputed with a different method (resampling the running total to Monday-ending weeks) and agree exactly.

2.2 Ebola, Guinea, Liberia and Sierra Leone, 2014

The Ebola file records cumulative cases (suspected, probable and confirmed) on irregular report dates from 22 March 2014 (day 0) to 5 January 2015, with 93 reports for Guinea, 83 for Liberia and 87 for Sierra Leone. Two problems were found and corrected.

Day numbers. The file's own day column disagrees with its date column by one day on 8 rows. Dates are the reliable field, so day numbers are recomputed as the number of days since 22 March 2014.

Running totals that fall. A cumulative count cannot decrease, but the reported totals fall 7 times for Guinea, 4 times for Liberia and 3 times for Sierra Leone, typically when suspected cases are later ruled out. Deleting the offending reports would discard data and depend on which report is blamed. Instead we replace the reported totals $y_1, \dots, y_n$ by the closest non-decreasing sequence in the least-squares sense, the *isotonic regression* [2]:

$$\hat{c} = \operatorname*{arg\,min}_{c_1 \le c_2 \le \cdots \le c_n} \sum_{i=1}^{n} (y_i - c_i)^2. \tag{2}$$

The solution is found by the pool-adjacent-violators algorithm: wherever a value is smaller than the one before it, the offending neighbours are pooled and replaced by their average, and pooling continues until no violation remains. For example, Guinea reported 1906, 1667, 1731 and 1760 cases on 27 and 29 October and 2 and 3 November 2014. The first value exceeds all three that follow, so all four are pooled and replaced by

$$\tfrac{1}{4}(1906 + 1667 + 1731 + 1760) = \tfrac{7064}{4} = 1766,$$

which is not above the next report. Where the data already rise, nothing changes. The final totals (2,776, 8,166 and 10,030) are unchanged, and once a country has passed 100 cases no value moves by more than 7.3% (Figure 2).

Weekly grid. Later stages need equally spaced observations. We evaluate the cleaned running total every 7 days by linear interpolation between neighbouring reports,

$$C(t) = \hat{c}_i + \frac{\hat{c}_{i+1} - \hat{c}_i}{d_{i+1} - d_i}\,(t - d_i), \qquad d_i \le t \le d_{i+1}, \tag{3}$$

where d_i is the day of report i, and take weekly differences. These weekly values are interpolated, not reported, counts; this matters for the choice of likelihood in Stage 5. The data end on 5 January 2015, so the series cover the rise and the first part of the decline, but not the end of the epidemic [13].

2.3 Measles, three US outbreaks, 2025–26

The measles series are those of our earlier two-group measles study and serve as an integer-order benchmark. They come from the Johns Hopkins University US Measles Data Repository [7]:

- *West Texas*: Texas weekly cases by rash-onset date, 19 January to 7 September 2025 (34 weeks, 802 cases);
- *Upstate South Carolina*: Spartanburg, Cherokee, Greenville and Pickens counties, by report date, 7 September 2025 to 26 April 2026 (34 weeks, 990 cases);
- *Short Creek (Utah–Arizona)*: Southwest Utah Health District and Mohave County, Arizona, by report date, 22 June 2025 to 31 May 2026 (50 weeks, 550 cases).

In the county file each row holds the new cases announced on one date, so a week's count is the sum over that Sunday-to-Saturday week. All three totals reproduce the earlier study exactly.

2.4 Take-off and reporting spikes

Two further rules, applied identically to all seven series, prepare the data for model fitting. Let $y_0, \ldots, y_K$ be the weekly counts of a series and k^* the week of its peak.

Definition 1 (Take-off week)**.** A *quiet run* is a block of at least three consecutive weeks with $y_k < 1$. The take-off week k_0 is the first week after the last quiet run that ends before k^*; if there is no such run, $k_0 = 0$.

The rule matters for this paper in particular. Liberia's series begins with a small cluster in late March 2014 followed by nine weeks without new cases; Sierra Leone's begins with seven nearly empty weeks. A model fitted from week 0 would read these flat stretches as very slow growth, which is precisely the kind of pattern that can push a fitted α below one. The rule gives take-off on 14 June 2014 for Liberia and 24 May 2014 for Sierra Leone, and week 0 for the other five series.

Definition 2 (Flagged week)**.** Week k is flagged as a possible reporting batch if

$$y_k > 3 \max\bigl\{\text{median}(y_{k-2}, y_{k-1}, y_{k+1}, y_{k+2}),\, 5\bigr\},$$

using whichever of the four neighbours exist. The floor of 5 cases prevents small-number noise from being flagged.

Seven weeks are flagged (circles in Figure 1). Three are clear reporting batches: Liberia's running total jumps from 4,665 to 6,535 between 18 and 25 October 2014 and then barely moves for a week, and both South Carolina and Short Creek report a holiday backlog in the week of 28 December 2025 (174 and 79 cases). The others (Guinea, 26 July 2014; Sierra Leone, 31 May 2014; Liberia, 29 November 2014; South Carolina, 15 March 2026) are ambiguous and include genuine take-offs. Flagged weeks are kept in the data; how they enter the likelihood is decided in Stage 5.

2.5 Summary

Table 2 summarises the seven series and Figure 1 shows them.

Table 2: The seven weekly series. "Total" is the sum of the weekly counts in the window (for Ebola, from the first full week of the interpolated grid). Peak week and take-off week are counted from week 0.

Series	Window	Weeks	Total	Peak (week)	Take-off	Dated by
Mpox, US	9 May – 19 Dec 2022	33	30,056	3,270 (12)	0	report
Ebola, Guinea	29 Mar 2014 – 3 Jan 2015	41	2,720	174 (28)	0	report
Ebola, Liberia	5 Apr 2014 – 3 Jan 2015	40	8,159	1,865† (29)	10	report
Ebola, Sierra Leone	5 Apr 2014 – 3 Jan 2015	40	9,720	941 (29)	7	report
Measles, W. Texas	19 Jan – 7 Sep 2025	34	802	84 (8)	0	rash onset
Measles, Upstate SC	7 Sep 2025 – 26 Apr 2026	34	990	174† (16)	0	report
Measles, Short Creek	22 Jun 2025 – 31 May 2026	50	550	79† (27)	0	report

† The peak week is a flagged reporting batch (Definition 2).

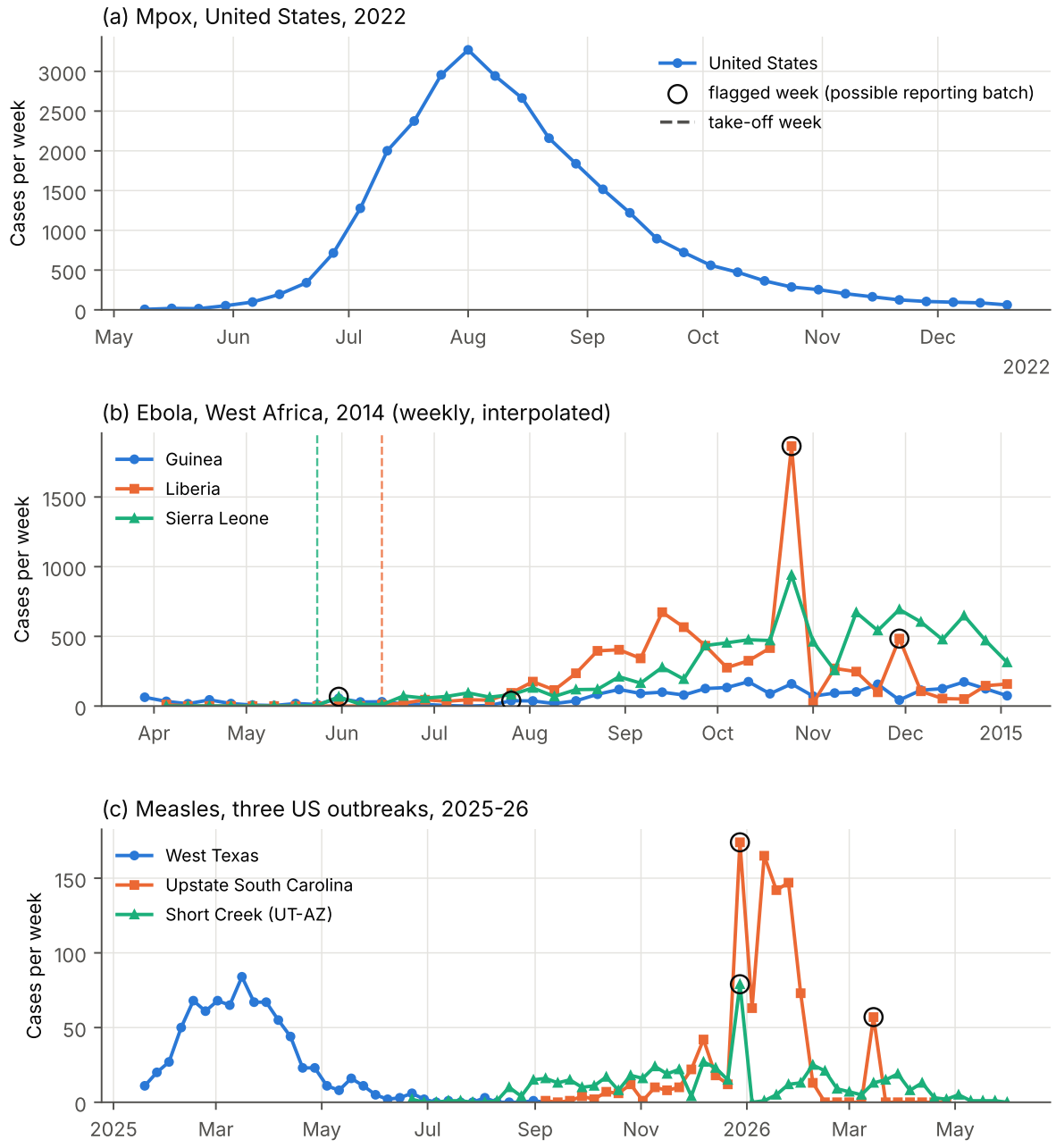


Figure 1: Weekly cases. (a) Mpox, United States, 2022. (b) Ebola, 2014, on the interpolated weekly grid. (c) Measles, three US outbreaks. Circles mark flagged weeks (Definition 2); dashed lines mark take-off weeks after week 0 (Definition 1).

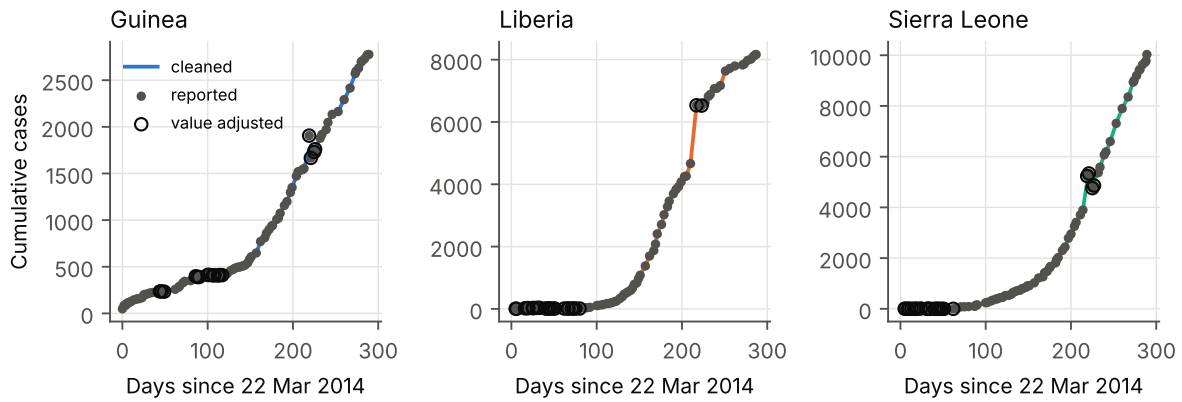


Figure 2: Ebola running totals: reported values (dots), values changed by the isotonic regression (2) (circles) and the cleaned, non-decreasing curve (line).

2.6 A first look at growth shape

Before fitting any model, Figure 3 plots cumulative cases from take-off to peak on two kinds of axes. If $C(t) = C_0 e^{rt}$ (exponential growth), then $\log C = \log C_0 + rt$ is a straight line against t (panel a). If $C(t) = At^p$ (power-law growth), then $\log C = \log A + p \log t$ is a straight line against $\log t$ (panel b). Fractional-order models produce power-law growth at early times, because the Caputo derivative of t^p is again a power of t.

No series is a single straight line in panel (a): every curve bends down, so the weekly growth rate falls over time. This is the sub-exponential pattern a fractional model can reproduce, and it is equally consistent with depletion of susceptibles, behaviour change and interventions. The figure therefore states the paper's question rather than answering it; Stages 4–6 test which explanation the data support.

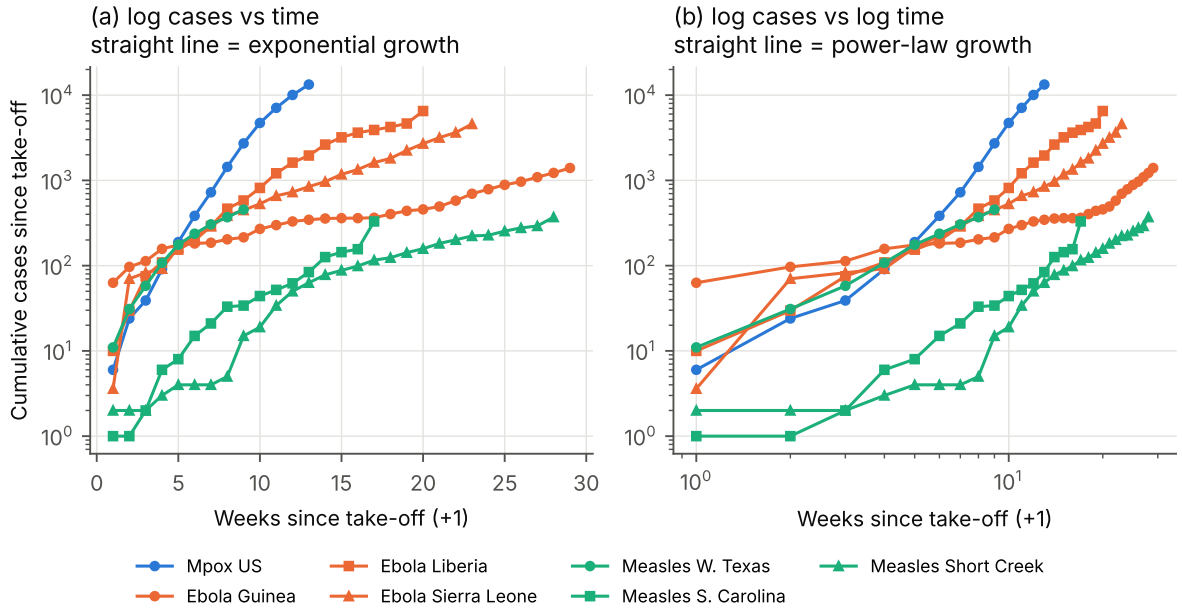


Figure 3: Cumulative cases from take-off to peak against weeks since take-off (plus one, so the first point is at $t = 1$). (a) Logarithmic vertical axis: exponential growth is a straight line. (b) Both axes logarithmic: power-law growth is a straight line.

3 Models and theory

[Planned, Stage 2.]

- The Caputo derivative in plain language and in symbols; the Mittag-Leffler function.
- Caputo fractional SEIR with dimensionally consistent rates $(\beta^\alpha, \sigma^\alpha, \gamma^\alpha)$ [5].
- The Angstmann–Henry–McGann fractional-recovery SIR model [1].
- Integer-order alternatives: Erlang (gamma-stage) SEIR, time-varying transmission $\beta(t)$, and a reporting-delay model.
- Existence, uniqueness and positivity; basic reproduction number; stability of the equilibria by Matignon's criterion [8].
- Proposition: what case data can and cannot identify (α versus transmission).

4 Numerical methods

[Planned, Stage 3.]

Garrappa's predictor–corrector method and the L1 scheme [6]; convergence test against a known Mittag-Leffler solution.

5 Synthetic experiments: does a memoryless process yield $\alpha < 1$?

[Planned, Stage 4.]

Simulate from the integer-order models of Section 3, fit the fractional model, and record the fitted α.

6 Fitting to outbreak data

[Planned, Stage 5.]

Likelihood for weekly counts (handling interpolated Ebola counts and flagged weeks), parameter estimates and profile likelihood for α.

7 Model comparison and forecasting

[Planned, Stage 6.]

AIC comparison and rolling-origin out-of-sample forecasts.

8 Discussion

[Planned, Stage 7.]

Data and code availability

All data are public (Table 1). The Python code will be made public on GitHub on submission.

Declarations

Funding: none. **Competing interests:** none.

A Reproducibility: Stage 1 scripts and checks

The four Stage 1 scripts run in order (Python 3.13, packages pinned in `requirements.txt`):

1. `01_download_data.py` downloads the four files from their pinned commits and verifies their SHA-256 fingerprints;
2. `02_clean_data.py` builds the weekly series, applies the isotonic cleaning, the take-off rule and the spike flags, and writes the summary in Table 2;
3. `03_plot_data.py` draws Figures 1–3;
4. `04_check_data.py` runs 12 independent checks.

The checks confirm that: (1) the raw files match their fingerprints; (2) all seven series are present with no missing values; (3) no weekly count is negative; (4) every series lies on a 7-day grid; (5) weekly counts add up to the running totals; (6) the mpox series agrees with an independent recomputation; (7) the measles totals reproduce the earlier study (802, 990, 550); (8) the cleaned Ebola totals never decrease; (9) past 100 cases, cleaning changes no Ebola value by more than 10% (largest change 7.3%); (10) the final Ebola totals are unchanged; (11) the weekly Ebola grid matches the cleaned reports on shared days; and (12) take-off precedes the peak in every series. All 12 checks pass. To confirm that the checks can fail, one measles count was altered by a single case; check (7) then failed as expected.

B Development log (remove before submission)

Stage 1, data (26 Oct 2026). Goal: reproducible, cleaned weekly series for all seven outbreaks. Done: pinned downloads with fingerprints; mpox weekly series from the running total; Ebola isotonic cleaning and weekly grid; measles series reproduced; take-off rule; spike flags; three figures; 12 checks passing. Important points carried forward: (i) fit Liberia and Sierra Leone from their take-off weeks; (ii) Ebola weekly counts are interpolated, so the likelihood should respect the reporting intervals or use cumulative counts; (iii) three clear reporting batches must be handled in the likelihood; (iv) every series bends down on a log scale, so slow growth is present in all of them and must be explained by one mechanism or another.

Next: Stage 2, models and theory. Fractional SEIR, AHM model, integer-order alternatives, proofs, identifiability proposition.

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